I Definitions

Def: A set of vectors $v_1, \dots, v_n$ are linearly independent if the linear combination of

$$a_1 v_1 + \dots + a_n v_n = 0$$

only when $a_1 = a_2 = \dots = a_n = 0$.

Def: The span of vectors $v_1, \dots, v_n$ is

$$\text{span}((v_1, \dots, v_n)) = \{ v_1 \in S : v = a_1 v_1 + \dots + a_n v_n \quad \forall\, a_1, a_2, \dots a_n \in \mathbb{R} \}$$

Def: $T = \{ v_1, v_2, \dots, v_n \}$ is a spanning set for a vector space V if

$$V = \text{span}(T)$$

Def: A basis of a vector space V is a spanning set that is linearly independent

Def: A dimension of a vector space V is the number of elements that form a basis for V

$$\dim(V) = |T| \quad \text{if } T \text{ is a basis set.}$$

A) Notation

Vectors $p_1, p_2, \dots, p_n$ where each $p_i \in \mathbb{R}^m$

Form matrix $A = \begin{bmatrix} p_1 & p_2 & \cdots & p_n \end{bmatrix}_{m \times n}$

Linear combination

$$\underline{x} = c_1 p_1 + c_2 p_2 + \dots + c_m p_m = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_m \end{bmatrix} \begin{bmatrix} c_1 & c_2, \dots, & c_m \end{bmatrix}$$

Def: Let V be a vector space. Then a real valued function

$$\|\cdot\| : V \to \mathbb{R}$$

is a norm if

1) $\|x\| \geq 0 \quad \forall x \in V$

2) $\|x\| = 0$ iff $x = 0$

3) $\|\alpha x\| = |\alpha| \, \|x\| \quad \forall \alpha \in \mathbb{R} \quad x \in V$

4) $\|x+y\| \leq \|x\| + \|y\|$ (triangle equality)

This looks very similar to a metric. Lets leverage that
For a metric space that is a vector space

$$d(x,y) = \|x-y\|$$

$$\|x\| = d(x,0)$$

B) Examples

$$\|x\|_1 = \sum_{i=1}^{d} |x_i| \quad l_1$$

$$\|x\|_p = \left[\sum_{i=1}^{p} |x_i|^p\right]^{1/p} \quad l_p$$

$$\|x\|_2 = \left[\sum_{i=1}^{d} |x_i|^2\right]^{1/2} \quad l_2$$

$$\|x\|_\infty = \max_{1 \leq i \leq d} |x_i| \quad l_\infty$$

Example

V = a space of functions

$\|x(t)\|_1 = \int_a^b |x(t)|\,dt$ $\quad L_1$: Set of functions $x(t)$ s.t. $\|x(t)\|_1 < \infty$

Most of our work will be here $\rightarrow$ $\|x(t)\|_2 = \left[\int_a^b |x(t)|^2\,dt\right]^{1/2}$ $\quad L_2$: Set of functions $x(t)$ s.t. $\|x(t)\|_2 < \infty$

$\|x(t)\|_p \left[\int_a^b |x(t)|^p\,dt\right]^{1/p}$ $\quad$ "

$\|x(t)\|_\infty = \sup_{t\in[a,b]} |x(t)|$ $\quad$ "

II Inner Product spaces

Def: Let V be a vector space. A function $\langle\,,\,\rangle : V \times V \to \mathbb{R}$ is an inner product if

1) $\langle x,y\rangle = \langle y,x\rangle$ (real)

$\langle x,y\rangle = \langle y,x\rangle^*$ (complex)

2) $\langle \alpha x,y\rangle = \alpha\langle x,y\rangle$

$\langle x,\alpha y\rangle = \alpha\langle x,y\rangle$ (real)

$= \alpha^*\langle x,y\rangle$ (complex)

3) $\langle x+y,z\rangle = \langle x,z\rangle + \langle y,z\rangle$

4) $\langle x,x\rangle > 0$ if $x \neq 0$

$\langle x,x\rangle = 0$ if $x = 0$

Examples

$$\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad \langle x, y \rangle = x_1 y_1 + \cdots + x_n y_n$$

$$= x^T y$$

$$x, y \in \mathbb{C}^n \quad \langle x, y \rangle = \sum_{i=1}^{n} x_i y_i^* = y^H x$$

$L_2[0,1]$ Real

$$x(t), y(t) \in L_2[0,1]$$

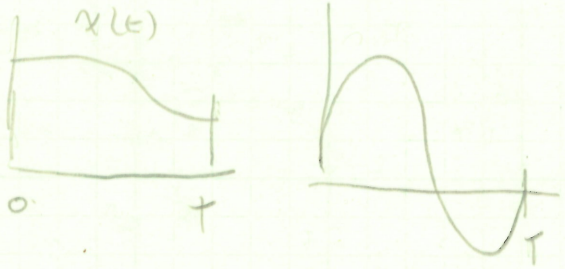
$$\langle x(t), y(t) \rangle = \left[\int_0^1 |x(t) y(t)|^2 dt \right]^{1/2}$$

$L_2[0,1]$ complex

$$\langle x(t), y(t) \rangle = \left[\int_0^1 |x(t) y^*(t)|^2 dt \right]^{1/2}$$

$L_2(-\infty, \infty)$

$$\langle x(t, e^{j\omega t} \rangle = \left[\int_{-\infty}^{\infty} |x(t) e^{-j\omega t}|^2 dt \right]^{1/2}$$



$$x(t) \longrightarrow \boxed{h(t)} \xrightarrow{y(t)} y(T)$$

$$y(T) = x(t) * h(t)\Big|_{t=T} = \int_0^T x(\tau)\, h(T-\tau)\, d\tau$$

Introduce $g(\tau) = h(T-\tau)$

$$y(T) = \int_0^T x(\tau)\, g(\tau)\, d\tau = \langle x(t), g(t) \rangle$$

$A, B \in \mathbb{R}^{m \times n}$

$$\langle A, B \rangle = tr(B^T A)$$

$A, B \in \mathbb{C}^{m \times n}$

$$\langle A, B \rangle = tr(B^H A)$$

III Induced Norm

$$\|x\|_2 = [\langle x, x \rangle]^{1/2}$$